Stefano **Lacovara**

Control Systems Engineer

Cento, FE, Italy

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Experience

Turintech — Metatronix *Turin, Italy*

Embedded Software Engineer *Apr 2026 – Present*

- Develop an L2 switch driver, integrate a PoE ASIC, and implement L3 networking on custom STM32H7 hardware running FreeRTOS with CMSIS.
- Implement REST APIs for web-based configuration and eMMC, I^2C, and GPIO firmware; bench-test connectivity, PoE operation, and configuration changes.

Università degli Studi di Padova — NAVLAB *Padova, Italy*

Research Fellow — NMPC Driving Copilot *Jun 2025 – Nov 2025*

- Adapted an existing autonomous-driving NMPC into a copilot by tuning weights and modifying C code.
- Implemented PID tracking of NMPC steering-angle references and maximum-demand pedal arbitration; conducted human-in-the-loop tests in VI-Grade's driving simulator.

New York University — Agile Robotics and Perception Lab *New York, USA*

Master's Thesis Research Intern *Apr 2024 – Oct 2024*

- Implemented a complete kinodynamic RRT* planner in C++/ROS to generate waypoints for an existing Bézier-curve trajectory optimizer for fixed-wing UAVs.
- Prepared UAVs and experiments, ran real-world flight tests, and checked planner operation.

Selected Project

Bosch Future Mobility Challenge *Cluj-Napoca, Romania*

Autonomous Driving Competition *Nov 2024 – Jun 2025*

- Integrated ToF, encoder, and line sensors with an STM32F4 microcontroller; tuned a PID controller for DC motor speed in a 1:10-scale autonomous vehicle.
- Developed embedded software in C/C++ with rosserial; tested the vehicle and competed in the finals.

Education

Università degli Studi di Padova *Padova, Italy*

M.Sc. in Control Systems Engineering *2023 – 2025*

- Grade: **110/110**. Thesis: Kinodynamic RRT* for path planning of a small fixed-wing UAV.

Università degli Studi di Ferrara *Ferrara, Italy*

B.Sc. in Electronic and Computer Engineering *2018 – 2022*

- Grade: **96/110**. Thesis: Analisi di Casi di Studio di BCI Impiegati per il Controllo.

Technical Skills

Software: C, C++, Python, MATLAB/Simulink, Bash, Git, Docker, CMake, TEX.
Control & planning: Control systems, optimal and nonlinear control, NMPC, PID, motion planning.
Embedded & firmware: STM32F4/H7, FreeRTOS, CMSIS, STM32CubeIDE, sensor integration.
Robotics & networking: ROS, rosserial, L2/L3 Ethernet switching, PoE.

Languages

Italian: Native. **English:** B2 upper-intermediate.